

Heart Disease Prediction Using Artificial Intelligence and Machine Learning

This project focuses on predicting heart disease using Machine Learning and Artificial Intelligence techniques. The analysis is performed using the provided spreadsheet dataset (**heart.csv**) and project requirements document. The system predicts whether a patient is likely to suffer from heart disease using medical parameters.

Problem Statement

Heart disease is one of the major causes of death globally. Early prediction can help doctors and healthcare professionals take preventive measures. Traditional diagnosis methods are expensive and time-consuming. This project develops an intelligent prediction system using Machine Learning models.

Objectives

- Predict heart disease using classification models.
- Compare multiple ML and AI algorithms.
- Evaluate model performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Identify important medical features affecting prediction.

Dataset Analysis

The provided dataset contains **303** patient records and **14** columns.

Important features include:

- age – Age of patient
- sex – Gender
- cp – Chest pain type
- chol – Cholesterol level
- trestbps – Resting blood pressure
- thalach – Maximum heart rate achieved
- exang – Exercise induced angina
- target – Heart disease presence

Data Preprocessing

The following preprocessing steps were performed:

- Checked for missing values
- Standardized numerical features
- Split dataset into training and testing sets
- Prepared data for ML and Neural Network models

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand feature distributions and relationships.

- Patients with higher chest pain levels showed higher disease probability.
- Cholesterol and heart rate showed strong influence.
- Age and exercise-induced angina also contributed significantly.

Machine Learning Models

- 1. **Logistic Regression** – Binary classification algorithm.
- 2. **Decision Tree** – Tree-based classification model.
- 3. **Random Forest** – Ensemble model combining multiple trees.
- 4. **Neural Network** – AI-based deep learning model capable of learning complex patterns.

Model Performance Evaluation

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	85.25%	87.1%	84.38%	85.71%	92.67%
Decision Tree	75.41%	84.0%	65.62%	73.68%	75.92%
Random Forest	83.61%	84.38%	84.38%	84.38%	92.03%
Neural Network	86.89%	90.0%	84.38%	87.1%	87.28%

Performance Analysis

The Neural Network model achieved the highest accuracy of **86.89%**. This indicates that AI-based methods can effectively learn complex medical patterns for heart disease prediction.

Random Forest also achieved strong performance with high stability and good ROC-AUC score.

Decision Tree showed comparatively lower accuracy due to overfitting tendencies.

Important Features Influencing Heart Disease:

- Chest Pain Type (cp)
- Maximum Heart Rate (thalach)
- Cholesterol Level (chol)
- Exercise Induced Angina (exang)
- Blood Pressure (trestbps)

Python Implementation

```
# Train Random Forest Classifier

from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

prediction = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, prediction))
```

Future Enhancements

- Integration with hospital systems
- Cloud deployment using Flask/Streamlit
- Real-time prediction using wearable devices
- Deep Learning optimization using larger datasets

Conclusion

This project successfully demonstrates the application of Artificial Intelligence and Machine Learning techniques in healthcare prediction systems. Using the provided dataset, multiple classification models were trained and evaluated. The Neural Network and Random Forest models achieved the best performance, making them suitable for predictive healthcare analytics and early heart disease detection.